Listing 4.10 Method contains

```
function contains(key,
positions=null)                                          #1
  if positions == null
then                                                    #2
    positions ← key2Positions(this.hashFunctions, this.seed,
key)             #3
  return positions.all((i) => readBit(this.bitsArray, i) !=
0)                #4
```

#1 Function `contains` takes a key and returns true if and only if all bits corresponding to the key are set to `1`. It is possible to pass explicitly the array of positions: some operations on the filter requires multiple accesses to a location, and this allows to save some computations. In languages allowing private methods and overloading, passing the second parameter should only be allowed for internal methods.

#2 Check if the array of positions has been passed: if so, avoid computing it again.

#3 Retrieves the bit indices corresponding to current key.

#4 Return true if and only if all bits read are not `0`. This line also uses a functional notation, with the `all` method that is similar to `map`, except it takes a predicate (a particular lambda function that returns a Boolean), applies it to every element in the list, and its result is true only if the predicate is true for all elements.